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Blueprint based exit material for

Fundamental database system

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❖ Database system, characteristics and application, different model of database

A database system is a collection of interrelated data that is organized for easy access, management, and use. It is a set of programs that enables users to create, maintain, and use a database.

The characteristics of a database system include:

- Real-world entity: A database system is designed to represent real-world entities and their relationships. For example, a database for a university might represent students, courses, and professors.
- Self-describing: A database system contains metadata, which is data about the data in the database. This metadata includes information about the structure of the database, the data types of the fields, and the relationships between tables.
- Data independence: A database system separates the data from the application programs that use the data. This means that changes to the data structure do not require changes to the application programs.
- Multiuser access: A database system allows multiple users to access the data at the same time. This is done by using a locking mechanism to ensure that only one user can update a particular piece of data at a time.
- Security: A database system must protect the data from unauthorized access, modification, or destruction. This is done by using security features such as passwords, encryption, and access control lists.

The applications of a database system include:

- Transaction processing: A database system can be used to process transactions, which are a sequence of operations that must be performed atomically. For example, a bank database system might be used to process a withdrawal transaction, which involves updating the account balance, debiting the account, and sending a notification to the customer.
- Data warehousing: A database system can be used to create a data warehouse, which is a repository of historical data. This data can be used for analysis, reporting, and decision-making.
- Business intelligence: A database system can be used to implement business intelligence (BI) solutions. BI solutions use data analysis and visualization techniques to help businesses make better decisions.

The different models of database include:

- Relational model: The relational model is the most common model for database systems. It uses tables to represent data, and relationships between tables are defined using foreign keys.
- Object-oriented model: The object-oriented model is a more recent model for database systems. It uses objects to represent data, and relationships between objects are defined using object-oriented techniques.
- NoSQL model: The NoSQL model is a newer model for database systems that is designed to handle large amounts of unstructured data. It does not use the traditional relational model, but instead uses different data models such as key-value pairs, documents, and graphs.

Each model has its own strengths and weaknesses, and the best model for a particular application will depend on the specific requirements of that application.

❖ Design ER models from specifications and interpret them into relational tables and write SQL statements data creation and manipulation

● SQL data creation

● SQL data manipulation

The SQL data manipulation language (DML) is used to query and modify database data. In this chapter, we will describe how to use the SELECT, INSERT, UPDATE, and DELETE SQL DML command statements, defined below.

- SELECT – to query data in the database
- INSERT – to insert data into a table
- UPDATE – to update data in a table
- DELETE – to delete data from a table

In the SQL DML statement:

- Each clause in a statement should begin on a new line.
- The beginning of each clause should line up with the beginning of other clauses.
- If a clause has several parts, they should appear on separate lines and be indented under the start of the clause to show the relationship.
- Upper case letters are used to represent reserved words.
- Lower case letters are used to represent user-defined words.

❖ Describe how to optimize databases and secure data

● Optimizing databases

- Indexing: Indexes are data structures that help speed up the retrieval of data from a database. By creating indexes on frequently used columns, you can improve the performance of queries that access those columns.
- Defragmentation: Over time, data in a database can become fragmented, which can slow down performance. Defragmenting the data can help to improve performance by grouping related data together.
- Caching: Caching is a technique that stores frequently used data in memory, which can improve performance by reducing the number of times the database needs to be accessed.
- Replication: Replication is a technique that creates copies of a database, which can improve performance by distributing the load across multiple servers.

● Securing data

- Passwords: Use strong passwords for all database users. Passwords should be at least 12 characters long and contain a mix of uppercase and lowercase letters, numbers, and symbols.
- Encryption: Encrypt sensitive data, such as credit card numbers and passwords. Encryption can help to protect data from unauthorized access.
- Access control: Implement access control to restrict who has access to the database. Access control can be implemented using roles and permissions.
- Backups: Regularly backup the database. This will help to protect data in case of a disaster.

By following these tips, you can optimize your databases and secure your data.

Here are some additional tips:

- Use the correct database for your application. For example, if you need to store a large amount of unstructured data, you might use a NoSQL database.
- Monitor your database performance. This will help you to identify any problems early on and take corrective action.
- Keep your database software up to date. This will help to protect your database from security vulnerabilities.